Paired-Acquisition Neural Factorization: An End-to-End Computational Pathology Pipeline

From Scanner-Aware Representations to Whole-Slide and Multi-Institutional Learning

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Preprint — 2026-09-04. Research use only; not clinical software.

Abstract. Paired-Acquisition Neural Factorization (PA-NF) is an end-to-end computational pathology research pipeline spanning representation learning, whole-slide neural aggregation, spatial modeling, and multi-institutional learning. The pipeline uses pretrained visual encoders where appropriate, while the paired-factorization methods, slide-level architectures, federated framework, institutional-weighting mechanisms, training and evaluation code, controlled stress experiments, and release infrastructure are developed within the program. The central question is how scanner, site, and institutional variation enters a pathology system and how that information changes as patches become representations, representations become slide predictions, and site-level models become collaborative models.

At the representation stage, the registered SCORPION campaign uses 48 human H&E slides, 480 aligned tissue regions, five scanners, and 2,400 images. Across 175 completed capacity-matched fits, PA-NF reduced tissue-branch scanner balanced accuracy by 0.3108 relative to an equal-capacity two-branch neural control, with fold/slide bootstrap interval $[-0.3346, -0.2858]$, while same-region retrieval remained within a preregistered 0.02 noninferiority margin and scanner information remained concentrated in an explicit acquisition branch. The frozen objective also transferred across DINOv2, Phikon, and ImageNet ResNet50 feature families. An independent five-scanner canine SCC study provides an external paired-scanner test and a harder boundary: PA-NF produces strong scanner separation, but the corrected fixed five-category comparison does not establish universal superiority over strong simple scanner-removal baselines.

At the whole-slide stage, 10,611 readable PANDA Phikon feature bags support TransnnMIL and federated experiments. Stabilized TransnnMIL remained competitive across 18 full-PANDA runs spanning six learning rates and three seeds, with mean best validation QWK 0.8257 at learning rate 10^{-4} and a best recorded run of 0.8455. At the institutional stage, a dominance-aware detector-switch policy on five simulated PANDA sites kept the clean regime essentially unchanged while improving global QWK by $+0.008009$ at 25% dominant-site corruption and $+0.007482$ at 35%. A fixed detector calibrated under label-noise stress transferred without retuning to systematic conservative ordinal threshold shift; at 45% shift it improved global QWK by $+0.01053$, macro-F1 by $+0.01512$, and worst-site QWK by $+0.01290$, with positive 95% intervals for all three metrics. CAMELYON17/WILDS adds a natural five-center test over 455,954 examples: equal-client weighting improved held-out center accuracy from 0.8312 to 0.9132 on frozen ImageNet features, while dominant-source downweighting reached 0.9322 with source-trained features versus 0.9052 for sample-proportional weighting. PCam provides a complete patch-level benchmark, with a documented run reaching 0.9394 ROC AUC, 0.8526 accuracy, and 0.8507 F1 on the official 32,768-patch test split.

Together, these studies define PA-NF as a connected computational pathology pipeline rather than a single representation model: scanner-aware paired representation learning feeds into whole-slide modeling and then into explicit studies of institutional influence, failure, and adaptive aggregation. The strongest conclusions are comparator-specific: PA-NF has a registered controlled advantage over its equal-capacity neural control on SCORPION; dominance-aware aggregation improves over FedAvg in specified PANDA stress regimes; and alternative source weighting improves held-out-center accuracy in the CAMELYON17 proxy. These results do not constitute clinical validation or a universal state-of-the-art claim.

1. Introduction

Computational pathology is often divided into separate model problems: patch classification, foundation-model feature extraction, multiple-instance learning, domain robustness, or federated learning. PA-NF is built from the opposite direction. It treats the pathology system as one connected pipeline in which acquisition and institutional effects can enter at several stages and can be amplified as information moves from patches to representations, from representations to slide predictions, and from local models to collaborative models.

The name *Paired-Acquisition Neural Factorization* originated in the representation stage, where exact same-tissue acquisitions expose what changes with scanner while holding the biological region fixed. That paired design remains the foundation of the program, but PA-NF now denotes the full pipeline built around it. Within this manuscript, *PA-NF representation factorization* refers specifically to the tissue/acquisition two-branch model, while *the PA-NF pipeline* refers to the complete computational pathology system.

The pipeline contains four principal authored systems:

- **PA-NF representation factorization:** a paired-scanner representation model with explicit tissue-oriented and acquisition-oriented branches, a joint decoder, crossed reconstruction, paired consistency, variance control, and scanner-routing objectives;
- **TransnnMIL:** a whole-slide multiple-instance learning architecture family combining transformer-style global correlation modeling with gated aggregation and optional hierarchical, topological, and pruning components;
- **PathologyFL:** a pathology-specific federated-learning framework with coordinator/client training, FedAvg/FedProx/FedAdam and custom weighting paths, robustness mechanisms, privacy integrations, monitoring, asynchronous execution, and failure handling;
- **FAIR-WEIGHTS-H and dominance-aware aggregation:** institutional weighting and detector-switch mechanisms that test when sample-count authority should yield to cross-site evidence, uncertainty, and reliability signals.

These systems share a common experimental layer. PCam provides patch-level training and evaluation. SCORPION and multi-scanner canine SCC provide exact paired acquisitions across five scanners. PANDA provides 10,611 readable slide-level Phikon feature bags for whole-slide modeling and simulated institutional stress. CAMELYON17/WILDS supplies 455,954 examples across five centers for natural center-shift studies. Controlled latent-factor generators provide ground truth for mechanism questions that cannot be observed directly in pathology data.

The central scientific hypothesis is that scanner and site variation should not be treated as a single preprocessing nuisance. A scanner shortcut can be encoded in a patch representation, attended to by a slide model, and then amplified through institutional aggregation. Conversely, institutional reweighting can change which representation failures dominate the global model. PA-NF therefore studies the transitions between these levels rather than optimizing them in isolation.

The main contributions of this paper are: (i) a registered paired-acquisition representation result on SCORPION with an equal-capacity neural control and cross-backbone transfer; (ii) an independent multi-scanner canine study that tests the limits of the representation claim against strong simple corrections; (iii) an implemented whole-slide architecture and full-PANDA stabilization study; (iv) a custom federated pathology stack with a dominance-aware detector that improves over FedAvg in controlled PANDA failure regimes and transfers without retuning to systematic ordinal shift; and (v) a CAMELYON17 held-out-center study showing that source influence and source sample count are not interchangeable.

The goal is not to collapse these experiments into one leaderboard. Each dataset answers a different question at a different level of the pipeline. The result is a single research system with several experimentally linked components, not a claim that one metric or one model dominates all of computational pathology.

2. The PA-NF pipeline

PA-NF is organized around three learned aggregation stages connected by a common pathology data layer. The same system can therefore be interrogated at the point where scanner information enters a representation, at the point where thousands of patch representations become a slide prediction, and at the point where site-level updates or source models are combined.

$$
\begin{aligned}
h_{r,s} &= E_\omega(x_{r,s}), \\
\mathscr{F}_{\text{pair}} :\ \{h_{r,s}\}_{s\in\mathscr{S}_r} &\longmapsto \left(z^{\text{tissue}}_{r,s}, z^{\text{acq}}_{r,s}\right), \\
\mathscr{A}_{\text{slide}} :\ \{(z_{k,j}, p_{k,j})\}_{j\in\mathscr{B}_k} &\longmapsto \widehat{y}_k, \\
\mathscr{A}_{\text{site}} :\ \{(\Delta\theta_i^{(t)}, w_i^{(t)})\}_{i\in\mathscr{I}_t} &\longmapsto \theta^{(t+1)}, \\
\mathscr{P}_{\text{PA-NF}} &= \mathscr{A}_{\text{site}} \circ \mathscr{A}_{\text{slide}} \circ \mathscr{F}_{\text{pair}} \circ E_\omega.
\end{aligned}
\tag{1}
$$

Here E_ω is a frozen or task-trained visual encoder, $\mathscr{F}_{\text{pair}}$ is the paired-acquisition representation stage, $\mathscr{A}_{\text{slide}}$ is whole-slide aggregation (TransnnMIL and matched MIL baselines), and $\mathscr{A}_{\text{site}}$ is institutional aggregation (PathologyFL, FedAvg-family baselines, and dominance-aware weighting). Spatial coordinates $p_{k,j}$ are optional inputs to the slide stage for spatial and topology experiments.

2.1 Stage I: paired representation factorization

For a tissue region r observed under scanner s, a visual encoder produces $h_{r,s}$. The PA-NF representation factorizer maps this feature into a tissue-oriented representation and an acquisition-oriented representation. Exact same-region multi-scanner correspondences provide the central supervisory structure. The main experiments use DINOv2 features, with frozen transfer to Phikon and ImageNet ResNet50. SCORPION provides the registered human paired-scanner experiment; multi-scanner canine SCC provides an independent external geometry with descriptive tissue categories.

2.2 Stage II: whole-slide aggregation

A whole slide is represented by a bag $\mathscr{B}_k$ of patch features, optionally paired with spatial coordinates. TransnnMIL combines a TransMIL-style global-correlation branch with an nnMIL-style gated aggregation branch and supports hierarchical spatial pooling, topology/GNN processing, branch-token fusion, graph caching, and controlled architectural variants. PANDA supplies the primary whole-slide substrate through 10,611 readable Phikon feature bags.

2.3 Stage III: institutional aggregation

PathologyFL extends the pipeline from a single slide model to simulated multi-site learning. The framework includes standard federated optimizers and custom institutional weighting, while the PANDA stress program tests a specific failure mode of sample-proportional aggregation: the largest site can retain the largest influence even when its label process becomes less aligned with the declared validation objective. CAMELYON17/WILDS provides a separate natural center-shift setting for studying source influence on an unseen center.

2.4 A shared experimental question

Across all three stages, the operative question is not simply whether a nuisance variable can be predicted. It is whether changing how information is allocated or weighted changes the downstream object that matters: same-region structure at the representation stage, slide prediction at the MIL stage, or held-out/site-level performance at the institutional stage. This is why scanner suppression, slide aggregation, and site weighting are treated as connected parts of one computational pathology pipeline.

The program uses strong conventional comparators at each stage: scanner-removal projections and paired-linear transforms for representation learning; mean pooling, AttentionMIL, TransMIL/nnMIL-family controls for whole-slide learning; and FedAvg-style sample weighting or equal-client weighting for institutional studies. Claims are therefore attached to specific comparisons rather than to a universal leaderboard.

3. Datasets and benchmark layer

The PA-NF pipeline has been exercised across paired-scanner, patch-level, whole-slide, and multi-center pathology settings. Table 1 summarizes the real-data substrates used in the program. Synthetic generators are used separately for

Table 1: Real pathology datasets used across the PA-NF pipeline. The datasets answer different questions and are not combined into one leaderboard.

Dataset	Scale used in this program	Pipeline stage	Primary endpoint	Role
PCam	Official 32,768-patch test split	Patch / engineering	ROC AUC, accuracy, F1	Full training/evaluation and federated smoke substrate
SCORPION	48 human H&E slides, 480 aligned regions, 5 scanners, 2,400 images	Representation	Scanner balanced accuracy, same-region retrieval	Registered paired-acquisition experiment
Multi-scanner canine SCC	44 biological samples; 805 geometry-qualified complete five-view regions across 5 scanners	Representation	Scanner recovery, descriptive category BA, retrieval	Independent paired-scanner external test
PANDA	10,611 readable Phikon slide feature bags	Whole-slide / institutional	QWK, macro-F1, worst-site QWK	MIL training, ordinal grading, simulated-site stress
CAMELYON17 / WILDS	455,954 examples from 5 centers	Institutional / center shift	Held-out-center accuracy, center recovery	Natural source-center weighting and center-subspace studies

mechanism and identifiability experiments.

3.1 PCam patch benchmark

PCam [8, 14] is the patch-level foundation. A ResNet18-based classifier with a transformer encoder/classification head was trained on the full training split and evaluated on the official 32,768-patch test split. The documented run reached ROC AUC 0.9394, accuracy 0.8526, and F1 0.8507. Patch-level bootstrap intervals were 0.9369–0.9418 for AUC and 0.8483–0.8563 for accuracy. This establishes a complete patch training/evaluation path and strong single-split discrimination; it is not used as a state-of-the-art or clinical-performance claim.

3.2 Feature substrates

The program deliberately separates the authored pipeline from the visual encoders used as inputs. PA-NF representation experiments use frozen DINOv2, pathology-native Phikon [6], and ImageNet ResNet representations. PANDA whole-slide experiments use 768-dimensional Phikon patch embeddings stored in slide-level HDF5 feature bags. CAMELYON17 experiments include both frozen ImageNet ResNet18 features and features from a source-trained CAMELYON17 ResNet18.

3.3 Statistical units

The inferential unit follows the data-generating structure. SCORPION is blocked by original slide; all scanner views and regions from a slide remain in the same fold. Canine SCC uses biological-sample-blocked folds. PANDA whole-slide analyses operate at the slide level. CAMELYON17 center studies preserve the WILDS center split, with one out-of-distribution validation center and one held-out test center. These units prevent the much larger counts of patches or scanner views from being treated as independent biological samples.

4. Stage I: paired-acquisition representation learning

$$
\begin{aligned}
h_{r,s} &= E_\omega(x_{r,s}), & b_{r,s} &= B_\theta(h_{r,s}), \qquad a_{r,s} = A_\phi(h_{r,s}),\\
\widehat{h}_{r,t|s} &= D_\psi(b_{r,s}, a_{r,t}), & &(r,s,t) \in \mathscr{M},\ s \neq t,\\
\mathscr{L}_{\text{PA-NF}} &= \lambda_{\text{xrec}} \sum_{(r,s,t)\in\mathscr{M}} \left\| D_\psi(b_{r,s}, a_{r,t}) - h_{r,t} \right\|_2^2 \\
&\quad + \lambda_{\text{pair}} \sum_{(r,s,t)\in\mathscr{M}} \|b_{r,s} - b_{r,t}\|_2^2 + \lambda_{\text{acq}}\, \mathrm{CE}(C_a(a_{r,s}), s) \\
&\quad + \lambda_{\text{dep}} \mathscr{R}_{\text{dep}}(b,a) + \lambda_{\text{var}} \mathscr{R}_{\text{var}}(b) + \lambda_{\text{proto}} \mathscr{R}_{\text{proto}}(b).
\end{aligned}
\tag{2}
$$

The representation stage asks whether matched acquisitions can provide a useful supervisory signal for allocating scanner-associated variation without simply erasing the representation. For a region observed under several scanners,

Table 2: Primary registered SCORPION comparison: PA-NF minus the equal-capacity two-branch neural control. Negative tissue scanner balanced accuracy is favorable; retrieval is judged against the registered 0.02 noninferiority margin.

Metric	Mean difference	95% fold/slide interval	Result
Tissue scanner balanced accuracy	-0.3108	$[-0.3346, -0.2858]$	Lower scanner recoverability
Mean same-region top-1 retrieval	$+0.00004$	$[-0.00009, +0.00026]$	Noninferior
Mean paired cosine	$+0.03639$	$[+0.03355, +0.03954]$	Positive geometry contrast
Worst paired cosine	$+0.03557$	$[+0.03167, +0.04064]$	Positive geometry contrast
Acquisition scanner balanced accuracy	$+0.1180$	$[+0.1003, +0.1358]$	Stronger explicit scanner routing

PA-NF learns a tissue-oriented branch $b_{r,s}$ and an acquisition-oriented branch $a_{r,s}$, with crossed reconstruction, same-region consistency, acquisition prediction, dependence control, and variance regularization. Tissue or diagnostic category labels are not required to construct the paired objective.

4.1 Registered SCORPION capacity-matched experiment

The primary human paired-scanner experiment uses SCORPION [12]: 48 original H&E slides, ten aligned regions per slide, and five scanner views per region, for 480 biological regions and 2,400 images. The registered capacity-matched campaign contains seven variants, five folds, and five seeds, for 175 completed fits. Its primary comparator is an equal-capacity two-branch neural control without the scanner-routing objectives.

The key result is the joint pattern: tissue-branch scanner recovery falls substantially relative to the capacity-matched neural control, same-region retrieval remains inside the predefined noninferiority margin, and scanner information becomes more recoverable from the acquisition branch. This supports a controlled comparative advantage on the registered SCORPION structured-separation objective. Because SCORPION does not supply a validated biological-category endpoint for these regions, the result is a representation result rather than a disease-biology claim.

4.2 Cross-backbone transfer

The frozen objective was transferred without backbone-specific tuning across DINOv2-Base, pathology-native Phikon, and ImageNet ResNet50 representations. In the transfer study, PA-NF reduced scanner recoverability in the tissue branch while retaining near-saturated same-region retrieval across all three feature families. Averaging the two unseen feature families within the same 48 SCORPION slides gave a scanner-probe difference of -0.4019 with interval $[-0.4145, -0.3895]$. The acquisition branch remained strongly scanner informative. This result shows that the paired objective is not tied to one visual encoder, although the registered DINOv2 capacity-matched campaign remains the primary controlled comparison.

4.3 Independent multi-scanner canine SCC study

The external paired-scanner study uses 44 canine cutaneous squamous-cell-carcinoma biological samples digitized on five scanners. After geometry qualification, 805 complete five-view regions are available. The corrected evaluation fixes five descriptive tissue categories—Dermis, Epidermis, Inflammation/Necrosis, SCC, and Subcutis—and uses biological-sample-blocked folds, fit-only standardization, and same-region/same-sample exclusions in nearest-neighbour candidate pools.

The canine experiment establishes an important boundary around the SCORPION result. PA-NF B32 attains the lowest mean scanner balanced accuracy in the corrected table, but strong centroid/QR and paired-linear corrections preserve higher retrieval while retaining similar descriptive category accuracy. Under the registered three-axis rule, neither B32 nor B64 establishes a universal neural feature-space increment over every simple baseline. Increasing the tissue bottleneck from 32 to 64 dimensions increases overall retrieval by $+0.0515$ and scanner recoverability by $+0.0489$, while corrected category balanced accuracy changes by -0.0076 with an interval crossing zero.

This is not a contradiction with the SCORPION controlled result. The two experiments use different comparators and endpoints. SCORPION establishes a bounded advantage over an equal-capacity neural control on the registered structured-separation objective; the canine study shows that broader scanner correction has a harder frontier in which

Table 3: Corrected five-fold canine SCC representation comparison. Category balanced accuracy is descriptive; lower scanner balanced accuracy and higher retrieval are favorable.

Representation	Category BA	Scanner BA	Overall retrieval	Worst-pair retrieval
Original frozen features	0.4439	0.8635	0.8724	0.7613
Centroid/QR scanner projection	0.4414	0.3278	0.9099	0.8196
Paired linear scanner transform	0.4422	0.3687	0.9022	0.8283
PCA component removal	0.3949	0.8157	0.9197	0.8512
PA-NF tissue branch B32	0.4331	0.2996	0.7192	0.5848
PA-NF tissue branch B64	0.4255	0.3486	0.7707	0.6345

strong linear methods remain competitive.

4.4 Pair integrity and information routing

Broken-pair controls provide a direct test of whether scanner suppression alone explains the result. Region- and sample-shuffled pairings can reduce scanner probe accuracy as much as, or more than, true same-region pairs, but they degrade same-region agreement and retrieval. Correct pairing therefore constrains what the system preserves rather than merely teaching it to confuse a scanner classifier.

The explicit acquisition branch is equally important. Scanner information remains strongly recoverable there, so the intended operation is allocation rather than wholesale deletion. The observed separation is partial and estimator-dependent: low linear scanner recoverability is not information-theoretic independence, and same-region retrieval is not a clinical or diagnostic endpoint.

5. Stage II: whole-slide neural aggregation

$$\begin{aligned}
&H_k = [h_{k,1}, \ldots, h_{k,n_k}]^\top, \\
&b_A = \mathscr{M}_{\text{Trans}}(H_k), \qquad && b_B = \mathscr{M}_{\text{att}}(H_k), \\
&T_k = \begin{bmatrix} P_A b_A \\ P_B b_B \end{bmatrix} \in \mathbb{R}^{2\times d}, \\
&Q_k = T_k W_Q, && K_k = T_k W_K, \qquad V_k = T_k W_V, \\
&A_k = \text{softmax}\left(\frac{Q_k K_k^\top}{\sqrt{d_h}} \right), \\
&\widetilde{T}_k = \text{LN}(T_k + A_k V_k), \\
&z_k = \frac{1}{2} \mathbf{1}^\top \widetilde{T}_k, && \widehat{y}_k = C_\eta(z_k).
\end{aligned} \tag{3}$$

$$\mathscr{B}_{\text{v2}} \subseteq \left\{ \mathscr{M}_{\text{Trans}}, \mathscr{M}_{\text{att}}, \mathscr{M}_{\text{hier}}, \mathscr{M}_{\text{graph}} \right\}, \qquad T_k = \text{stack}(\{P_j b_j : j \in \mathscr{B}_{\text{v2}}\}). \tag{4}$$

The second stage of PA-NF converts patch-level representations into slide-level predictions. A whole-slide image can contain tens of thousands of candidate tissue patches, so the learning problem is not only feature quality but how information is selected, combined, and propagated across the bag. TransnnMIL is the custom architecture family developed for this stage.

5.1 TransnnMIL architecture

The canonical implementation combines two complementary aggregation paths. A TransMIL-style branch models global correlations among patch embeddings [13], while an nnMIL/attention-style branch learns gated instance aggregation [7]. Their slide representations are projected into explicit branch tokens and fused with self-attention over the branch-token sequence. This corrected design allows genuine interaction between the branches rather than reducing to a

Table 4: Representative PANDA whole-slide results on Phikon feature bags. QWK is quadratic weighted kappa.

Model / run	Validation QWK
Mean-pooled Phikon + MLP	0.7274
Gated AttentionMIL	0.8100
TransnnMIL tuned seed 42	0.8155
TransnnMIL tuned seed 123	0.8225
TransnnMIL tuned seed 2025	0.8086

Table 5: TransnnMIL optimization-stability grid on the full readable PANDA feature manifest.

Learning rate	Mean best val QWK	Std.	Min.	Max.
1×10^{-4}	**0.8257**	0.0169	0.8087	0.8425
2×10^{-4}	0.8245	0.0192	0.8077	**0.8455**
3×10^{-4}	0.8238	0.0160	0.8127	0.8422
5×10^{-4}	0.8158	0.0144	0.8042	0.8319
7×10^{-4}	0.8117	0.0163	0.7994	0.8301
1×10^{-3}	0.8160	0.0170	0.7998	0.8337

single-position attention operation.

The wider TransnnMIL family also contains:

- hierarchical spatial pooling over learned regions;
- an optional topology branch with GATv2, GraphSAGE, or GIN processing;
- coordinate-aware k-nearest-neighbour graph construction and HDF5 graph caching;
- adaptive patch-pruning experiments;
- concat, gate, and learned branch-attention fusion controls; and
- two-branch and three-branch variants for matched ablation studies.

These modules are part of the authored whole-slide architecture. The headline empirical evidence below concerns the repaired canonical PANDA path; optional branches are treated as extensions unless they have their own matched real-data result.

5.2 Full-PANDA feature pipeline

PANDA [5] provides the primary slide-level substrate. The feature manifest was verified before training: 10,614 candidate HDF5 files were attempted, 10,611 were readable, and three unreadable files were removed. The resulting split contains 8,488 training slides and 2,123 validation slides, with 768-dimensional Phikon patch embeddings.

The baseline hierarchy is informative even before matched architecture-superiority testing. Mean-pooled Phikon followed by an MLP reached validation QWK 0.7274, while gated AttentionMIL reached 0.8100. Three tuned TransnnMIL runs reached 0.8155, 0.8225, and 0.8086.

5.3 Optimizer stabilization across 18 full-PANDA runs

Initial TransnnMIL experiments exposed substantial learning-rate sensitivity. The stabilized recipe therefore combined AdamW, warmup-cosine scheduling, two warmup epochs, gradient clipping at norm 1.0, and early stopping. Six learning rates were evaluated across seeds 42, 123, and 2025, producing 18 full-PANDA runs.

The principal whole-slide conclusion is therefore stability rather than field-wide superiority: a careful training recipe keeps the repaired canonical TransnnMIL competitive across a broad optimization range. A fully matched comparison against AttentionMIL, TransMIL, nnMIL, concat, gate, and learned branch-attention controls remains the correct test for an architecture-superiority claim.

5.4 Spatial tissue-dynamics extension

PA-NF also contains an experimental WSI-NCA line that replaces one-shot bag aggregation with repeated local state updates over spatial neighbourhoods. A frozen PANDA-300 ladder compared static, one-step, four-step tied, shuffled-topology, and untied controls across five seeds. The four-step tied model did not outperform the static or one-step controls on mean held-out QWK, and real topology did not consistently outperform shuffled topology. WSI-NCA therefore remains an active spatial-modeling extension rather than a demonstrated predictive improvement in the current pipeline.

6. Stage III: multi-institutional learning with PathologyFL

$$
\begin{aligned}
\Delta\theta_i^{(t)} &= \mathrm{LocalTrain}\Big(\theta^{(t)}, \mathscr{D}_i; E_i, \mu_i\Big), \\
a_i^{(t)} &= \mathbf{1}\Big[\mathrm{finite}(\Delta\theta_i^{(t)}) \wedge \tau_i^{(t)} \le T_i^{(t)} \wedge \neg\,\mathrm{Byz}_i^{(t)}\Big], \\
\widetilde{w}_i^{(t)} &= a_i^{(t)} w_i^{(t)} e^{-\gamma\tau_i^{(t)}}, \qquad \bar{w}_i^{(t)} = \frac{\widetilde{w}_i^{(t)}}{\sum_{j\in\mathscr{I}_t} \widetilde{w}_j^{(t)}}, \\
\theta^{(t+1)} &= \theta^{(t)} + \sum_{i\in\mathscr{I}_t} \bar{w}_i^{(t)} \Delta\theta_i^{(t)}.
\end{aligned}
\tag{5}
$$

The institutional weight $w_i^{(t)}$ can instantiate sample-proportional FedAvg, equal-client weighting, or a bounded adaptive policy such as FAIR-WEIGHTS-H. The availability gate $a_i^{(t)}$ captures finite-update, timeout, and robust-aggregation checks, while the exponential term represents optional staleness weighting in asynchronous execution.

The third stage of PA-NF studies how site-level information is aggregated. Standard FedAvg assigns influence in proportion to sample count [11]; this is appropriate only when sample volume is a reasonable proxy for the reliability and task alignment of a client's update. PathologyFL is the custom federated computational pathology framework developed to test that assumption and alternatives to it.

6.1 Framework architecture

PathologyFL implements a coordinator/client system with standard and custom aggregation paths. The framework includes:

- FedAvg, FedProx [10], FedAdam, explicit weighted aggregation, and Byzantine-robust variants;
- local pathology training with synchronous and asynchronous round execution;
- dropout, retry, timeout, checkpoint, and model-registry mechanisms;
- differential-privacy integrations and privacy-budget tracking [1];
- TLS/gRPC communication and secure-aggregation integrations [4];
- update anomaly and Byzantine-detection components, including Krum-family and robust summary operators [3];
- compression, resource-awareness, monitoring, and round-level instrumentation; and
- FAIR-WEIGHTS-H and dominance-aware weighting paths for experiments in which sample volume and site reliability diverge.

The implementation is a research framework rather than a deployed hospital federation. Its scientific role in this paper is to make institutional influence an explicit variable that can be perturbed and measured on pathology-derived data.

6.2 The dominant-site failure model

PANDA Phikon slide representations are partitioned into five simulated sites with one largest client. Controlled stress is then applied only to that dominant site's label process while the aggregation rule initially continues to assign influence by sample count. The design isolates a concrete failure mode: the largest site remains the most influential even after its training signal becomes less aligned with the validation objective.

The first intervention is cross-site blending, which reduces dominant-site authority. It is not universally better than

FedAvg: under clean conditions FedAvg retains advantages on some metrics. Under dominant-site corruption, however, cross-site blending significantly improves global QWK at 25% and 35% label noise and worst-site QWK at 45%. This motivates a conditional rather than permanent replacement of FedAvg.

6.3 Dominance-aware detector switching

The detector uses only observable validation behavior from the current FedAvg solution: global QWK, worst-site QWK, site-QWK spread, mean absolute ordinal error, and severe-error rate. Thresholds are calibrated from clean runs. The selected rule switches only when at least three diagnostics leave their clean-calibrated ranges, allowing the system to preserve FedAvg when its validation behavior remains normal and to reduce sample-size dominance when several independent indicators degrade.

This mechanism is evaluated in two distinct stress families: random dominant-site label corruption and systematic conservative ordinal threshold shift. The second experiment freezes the detector calibrated in the first family and applies it without retuning, providing a direct test of whether the decision rule transfers beyond the stress mechanism on which it was selected.

7. Institutional weighting beyond sample count

$$
\begin{aligned}
s_i^{(t)} &= \begin{cases} -\infty, & g_i^{(t)} = 0, \\ \alpha q_i^{(t)} + \beta u_i^{(t)} - \lambda p_i^{(t)}, & g_i^{(t)} = 1, \end{cases} \\
\pi_i^{(t)} &= \frac{\exp\left(s_i^{(t)}/\tau\right)}{\sum_{j=1}^{m} \exp\left(s_j^{(t)}/\tau\right)}, \\
w^{(t)} &= \Pi_{\mathscr{W}}\left((1-\eta)w^{(t-1)} + \eta \pi^{(t)}\right), \\
\mathscr{W} &= \left\{ w \in \mathbb{R}^m : \mathbf{1}^\top w = 1, \quad w_{\min} \le w_i \le w_{\max} \right\}, \\
\theta^{(t+1)} &= \theta^{(t)} + \sum_{i=1}^{m} w_i^{(t)} \Delta\theta_i^{(t)}, \\
\mathscr{H}_{\text{norm}}(w) &= -\frac{1}{\log m} \sum_{i=1}^{m} w_i \log w_i, \qquad N_{\text{eff}}(w) = \left(\sum_{i=1}^{m} w_i^2 \right)^{-1}.
\end{aligned}
\tag{6}
$$

FAIR-WEIGHTS-H is the institutional-weighting component of PathologyFL. Its purpose is to study aggregation rules in which influence can depend on more than local sample volume. The implemented engine combines adjusted validation quality, useful uniqueness, uncertainty penalties, integrity gates, bounded weights, entropy diagnostics, and conservative fallbacks.

For institution i, let q_i denote adjusted validation quality, u_i a useful-uniqueness signal, p_i an uncertainty penalty, and $g_i \in \{0,1\}$ an integrity gate. The implemented score takes the form

$$
s_i = \begin{cases} -\infty, & g_i = 0, \\ \alpha q_i + \beta u_i - \lambda p_i, & g_i = 1, \end{cases}
$$

with nonnegative coefficients α, β, λ. Scores are converted to normalized weights with either a softmax or a bounded mirror-descent update. Lower and upper weight caps prevent a single source from collapsing the aggregate, while normalized entropy and effective-client-count summaries quantify concentration of influence.

The wider design also specifies training/validation/monitoring weight separation, difficulty adjustment, subgroup constraints, contribution estimators, and explicit weight-change limits. Some of those extensions remain specification-level rather than fully implemented. Accordingly, the empirical claims in this paper come from the concrete dominance-aware PANDA experiments and the CAMELYON17 source-weighting studies, not from the method name itself or from a general fairness claim.

Table 6: Tuned dominance-aware detector on the full readable PANDA feature cache across 15 seeds. Deltas are detector-switch minus FedAvg.

Dominant-site corruption	Metric	FedAvg mean	Detector mean	Delta (95% interval)
0%	Global QWK	0.720137	0.720262	$+0.000126$ $[-0.000144, 0.000395]$
25%	Global QWK	0.694833	0.702842	$+0.008009$ $[0.001345, 0.014672]$
35%	Global QWK	0.689265	0.696747	$+0.007482$ $[0.003919, 0.011046]$
45%	Worst-site QWK	0.633666	0.643379	$+0.009713$ $[-0.000257, 0.019683]$

The scientific distinction is simple: *sample count is one signal for assigning influence, not a proof that the corresponding update is the most useful for the declared objective*. The following PANDA and CAMELYON17 studies test that statement under controlled and natural center-shift settings.

8. PANDA whole-slide and institutional results

PANDA [5] connects the whole-slide and institutional stages of PA-NF. The same 10,611-slide readable Phikon cache supports slide-level ISUP grading, model aggregation studies, dominant-site stress, and detector-transfer experiments.

8.1 Whole-slide model hierarchy

The slide-level hierarchy shows the gain from learned aggregation over a simple pooled baseline in this feature setting. Mean-pooled Phikon plus an MLP reached QWK 0.7274; gated AttentionMIL reached 0.8100; tuned TransnnMIL runs reached 0.8155, 0.8225, and 0.8086. The separate 18-run stabilization grid is reported in Section 5. These values establish a working and stable custom whole-slide architecture, while a definitive architecture-superiority claim awaits a fully matched comparison against all MIL controls.

8.2 Dominant-site label-noise stress

The full-cache institutional experiment uses five simulated sites and 15 seeds. Label noise is applied only to the largest site at 0%, 25%, 35%, and 45% corruption. Cross-site blending is intentionally not assumed to be universally preferable: in the clean regime, FedAvg retains an advantage on some site-level metrics. Under corruption, however, reducing dominant-site authority becomes beneficial.

The tuned observable detector requires at least three FedAvg validation diagnostics to leave their clean-calibrated ranges. This reduced the clean false-switch rate from 33.3% for the initial detector to 6.7%, while trigger rates remained 86.7%, 86.7%, and 93.3% at 25%, 35%, and 45% corruption respectively.

The result is conditional rather than universal: the detector is near-neutral when FedAvg remains well behaved and produces statistically positive global-QWK gains at 25% and 35% dominant-site corruption.

8.3 Transfer to systematic ordinal threshold shift

A stronger test freezes the detector selected under random dominant-site label noise and applies it without retuning to conservative ordinal threshold shift. This changes the failure mechanism from stochastic label corruption to a systematic shift in ordinal grading thresholds.

At 35% shift, the 95% intervals are $[0.00062, 0.01022]$ for global QWK, $[0.00272, 0.01405]$ for macro-F1, and $[0.00169, 0.01813]$ for worst-site QWK. At 45% shift, the corresponding intervals are $[0.00239, 0.01866]$, $[0.00819, 0.02204]$, and $[0.00547, 0.02034]$. The fixed detector therefore transfers across stress mechanisms at the stronger shift levels without being tuned on the transfer condition.

Leave-one-diagnostic-family ablation and a 36-setting calibration-neighbourhood sweep further test whether the result depends on one exact feature or one exact threshold choice. Removing the most frequent diagnostic family retains positive mean 35/45-shift deltas, and 29 of 36 nearby detector settings preserve low clean switching together with

Table 7: Fixed detector transfer from label-noise calibration to conservative ordinal threshold shift. Deltas are detector-switch minus the clean-strategy baseline.

Shift	Trigger rate	Global QWK Δ	Macro-F1 Δ	Worst-site QWK Δ
0%	13.3%	−0.00025	+0.00009	+0.00151
25%	33.3%	+0.00129	+0.00290	+0.00353
35%	60.0%	+0.00542	+0.00838	+0.00991
45%	73.3%	+0.01053	+0.01512	+0.01290

Table 8: Held-out center-2 accuracy under two feature families and three source-weighting policies.

Feature extractor	Sample-proportional	Equal-client	Downweight dominant
Frozen ImageNet ResNet18	0.8312	**0.9132**	0.9094
CAMELYON17-trained ResNet18	0.9052	0.9318	**0.9322**

positive 35%/45% gains across the three headline metrics.

8.4 Centralized, federated, and local-only ordering

A separate cached-feature comparison provides context for the cost of federation. On the 3,000-slide PANDA subset, global QWK followed the ordering centralized > FedAvg > local-only, with values 0.6949, 0.6659, and 0.6075 respectively. This is a simulated-site feature-level experiment rather than a claim about real hospital deployment, but it establishes that the PathologyFL experiments operate in a regime where institutional aggregation materially affects predictive performance.

9. CAMELYON17 natural center-shift studies

CAMELYON17/WILDS [2, 9] provides a complementary institutional setting in which center identity is observed rather than synthetically assigned. The feature-level dataset used here contains 455,954 labeled examples from five centers. Centers 0, 3, and 4 form the source-training pool, center 1 is used for out-of-distribution validation, and center 2 is held out for final testing.

9.1 Source weighting and held-out-center generalization

Three source-influence policies are compared: sample-proportional weighting, equal-client weighting, and downweighting the dominant source. The experiment is implemented as a centralized feature-level proxy for institutional influence, not as a full federated training run.

On frozen ImageNet features, equal-client weighting improves held-out-center accuracy by 8.20 percentage points over sample-proportional weighting. With source-trained CAMELYON17 features, the gap narrows but remains positive: dominant-source downweighting reaches 0.9322 versus 0.9052 for sample-proportional weighting. The source-trained sample-proportional model reaches approximately 0.9991 source-training accuracy, demonstrating that stronger source fit does not guarantee stronger performance on the unseen center.

These experiments reinforce the PANDA stress result from a different direction. More source samples can improve estimation when source and target distributions align, but sample volume alone does not determine which weighting best generalizes to a new center.

9.2 Center subspace diagnostics

The same dataset is used to study whether center identity occupies a partially removable representation subspace. Adversarial and subtractive nuisance-removal variants did not materially reduce post-hoc center leakage. A supervised linear center-subspace projection provided a cleaner mechanism test: five-center decodability fell from approximately

0.895 to 0.764 at effective rank four while tumor AUC remained approximately 0.990. This is evidence that part of the center signal is linearly structured and removable without a large loss on the measured tumor endpoint; it is not evidence of complete center invariance.

9.3 Communication and systems studies

The CAMELYON17 feature/head setting also provides a bounded systems calculation for federated communication. Across 100 fp32 rounds and three clients, exchanging a full ResNet18 was estimated at approximately 24.98 GB, compared with approximately 2.35 MB for a 512-to-2 classification head—a ratio of roughly 10,894. A coefficient-noise perturbation study preserved the equal-client and dominant-source-downweighting advantages through noise standard deviation 0.20, while a separate infrastructure simulation measured client-speed heterogeneity, dropout, failed rounds, and synchronous straggler burden.

These experiments are useful engineering bounds around the PA-NF institutional stage. They do not substitute for a real multi-hospital federated deployment, but they connect source weighting, center shift, communication cost, and infrastructure heterogeneity within the same research pipeline.

10. Controlled mechanism studies

Real pathology data do not reveal the true latent biological and acquisition factors. Controlled generators therefore complement the empirical pipeline by making those factors known and allowing recovery to be measured directly. These experiments are used to study mechanism and resource allocation, not to replace real pathology validation.

10.1 Observations, paired anchors, and repeated exposure

The synthetic program separates three resources: ordinary observations N_u, unique paired anchors N_p, and repeated paired presentations. Ordinary observations define the task manifold; unique anchors expose more biological locations under acquisition changes; repetition supplies additional optimization signal for already observed interventions.

In a matched-budget high-overlap nonlinear study, total paired exposure produced the largest effect. Doubling paired presentations from 6,400 to 12,800 increased the universal biological recovery score by $+0.037357$, with interval $[0.030315, 0.043657]$, positive in all ten seed blocks. Under 6,400 presentations, increasing unique anchors from 50 to 200 improved recovery by $+0.017865$ with interval $[0.010560, 0.026079]$, while the incremental 100-to-200 gain was only $+0.001578$. The same diminishing-return pattern appeared under the larger presentation budget.

The practical conclusion within this generator is that paired supervision has two separable resources: exposure and diversity. More pair-loss updates matter, but repeatedly observing only a very small anchor set is not equivalent to seeing a broader set of paired biological locations.

10.2 Capacity allocation

Factorial experiments vary tissue-branch dimensionality and cross-factor regularization to test how capacity moves information between branches. The real canine B32/B64 comparison mirrors one synthetic mechanism: increasing tissue capacity can recover more region identity while also allowing more scanner information through. In the corrected canine experiment, however, that capacity increase does not produce a corresponding descriptive category gain. Synthetic accessibility trends are therefore treated as mechanistic hypotheses rather than as biological conclusions.

10.3 Why synthetic experiments remain separate

The controlled generators establish properties of the specified observation models, mixing functions, optimization budgets, and evaluation metrics. They cannot establish scanner-free biology, clinical validity, or universal identifiability in histopathology. Their role in PA-NF is to explain why particular real-data experiments are worth running and to identify variables—pair exposure, anchor diversity, bottleneck width, overlap, and nonlinear mixing—that can be manipulated systematically.

11. Discussion

11.1 PA-NF as a connected pathology system

The main contribution of this work is not one isolated model but a pipeline in which acquisition, slide-level, and institutional effects can be studied in sequence. The SCORPION experiments show that same-tissue paired acquisitions can change how scanner information is allocated in a frozen representation. The PANDA experiments then move from representations to slide predictions and simulated multi-site aggregation. CAMELYON17 provides a separate natural center-shift setting in which source influence can be tested against an unseen center. Taken together, these studies support the premise that scanner and site effects are not merely preprocessing concerns: they can interact with later aggregation choices.

11.2 What the representation result establishes

The strongest representation result is comparator-specific and reproducible across folds: PA-NF has a registered controlled advantage over an equal-capacity two-branch neural control on the SCORPION structured-separation objective. The method reduces linear scanner recoverability in the tissue branch while satisfying the registered retrieval noninferiority requirement and increasing scanner routing to the acquisition branch. Cross-backbone transfer shows that this behavior is not confined to DINOv2.

The canine SCC result is valuable because it prevents that claim from becoming broader than the evidence. Strong centroid/QR and paired-linear corrections can occupy competitive points on the scanner/category/retrieval frontier. PA-NF therefore should not be described as the universally best scanner-removal method. Its demonstrated advantage is the registered SCORPION neural-control comparison together with an explicit factor allocation that remains useful for mechanism studies.

11.3 Whole-slide modeling is now a real pipeline stage

TransnnMIL is no longer a design sketch: it has a verified full-PANDA feature pipeline and an 18-run stabilization study. The most defensible current statement is that the repaired canonical architecture can be trained stably over a broad learning-rate range and produces competitive validation QWK in this setting. A matched architecture-superiority experiment remains necessary before claiming that the TransnnMIL fusion itself beats AttentionMIL, TransMIL, or nnMIL.

The negative WSI-NCA PANDA-300 ladder is also informative. Repeated spatial updates did not produce a predictive advantage in that frozen experiment, which argues against promoting spatial recurrence merely because it is architecturally appealing. Future spatial additions to PA-NF should earn their place by improving a controlled pathology endpoint.

11.4 Institutional influence is not equivalent to sample volume

The strongest institutional result is the detector-transfer experiment. In the clean PANDA regime the detector switches rarely and has near-zero global-QWK cost. Under stronger dominant-site misalignment, the same fixed rule increasingly switches away from sample-size dominance and improves global QWK, macro-F1, and worst-site QWK. The fact that a rule selected under random label-noise stress transfers without retuning to systematic conservative ordinal shift makes the result more informative than a separately optimized detector for each failure mode.

CAMELYON17 independently supports the same conceptual distinction. Sample-proportional weighting is not the best held-out-center policy in either tested feature family, even when the source-trained model fits the source data almost perfectly. These experiments do not prove a universal institutional weighting law, but they demonstrate that source quantity and target generalization can diverge.

11.5 Pipeline-level implication

The common thread is conditional allocation. At the representation stage, scanner-associated information should not simply be deleted if it overlaps with tissue information; it can be routed into an explicit acquisition branch. At the institutional stage, a large site's influence should not simply be removed because it is large; influence can be reduced when multiple validation signals indicate that sample-volume authority has become misaligned with the task. The same design principle appears at two different scales: preserve useful information while changing where and how unreliable variation is allowed to dominate.

This perspective makes PA-NF extensible. New spatial, representation, or optimization systems should enter the flagship pipeline only after they are tested against the existing stages and controls. That keeps the pipeline cumulative without treating every experimental branch as an established contribution.

12. Limitations

The present results remain bounded in several important ways.

- **Paired-scanner representation learning.** SCORPION contains 48 original human slides, and the independent labeled paired-scanner cohort is canine rather than human. The experiments use frozen feature representations rather than end-to-end pixel-space adaptation. Linear probe accuracy measures recoverability under a specified estimator, not information-theoretic independence, and same-region retrieval is not a diagnostic endpoint.
- **Whole-slide learning.** TransnnMIL has a full-PANDA stabilization result but not yet a fully matched architecture-superiority study against every relevant MIL control under identical tuning and compute. Optional hierarchical, topology, and pruning branches should not be interpreted as validated improvements until separately tested.
- **Spatial dynamics.** The current WSI-NCA PANDA-300 experiment does not support a recurrence- or topology-dependent predictive advantage. Spatial recurrence remains an experimental extension rather than an established benefit.
- **Federated learning.** PANDA sites are simulated partitions over pathology-derived feature data, not independent hospitals. The dominance-aware detector is validated under controlled corruption and ordinal-shift mechanisms; real institutional deployment would require prospective multi-center testing.
- **CAMELYON17.** The source-weighting studies are centralized feature-level proxies for institutional influence and use one held-out test center. They do not validate a complete federated algorithm or establish a general law across hospitals.
- **PCam and clinical interpretation.** PCam is a patch-level single-split benchmark, and none of the reported studies constitutes clinical validation, deployment guidance, or evidence of improved patient outcomes.

The pipeline is therefore best interpreted as a research system with several strong controlled results and several active extensions, rather than as a clinically validated end-to-end product.

13. Conclusion

Paired-Acquisition Neural Factorization is presented here as a complete computational pathology research pipeline extending from scanner-aware representations to whole-slide and multi-institutional learning. The representation stage uses same-tissue multi-scanner observations to route acquisition information between explicit tissue-oriented and acquisition-oriented branches. The whole-slide stage uses TransnnMIL and matched MIL baselines over a verified 10,611-slide PANDA feature pipeline. The institutional stage uses PathologyFL, dominance-aware aggregation, and center-weighting experiments to study how site influence changes when sample volume and task alignment diverge.

The strongest results are specific and complementary. On SCORPION, PA-NF has a registered controlled advantage over an equal-capacity two-branch neural control, reducing tissue-branch scanner balanced accuracy by 0.3108 while satisfying the predefined same-region retrieval noninferiority criterion and retaining scanner information in the acquisition branch. On PANDA, a fixed dominance-aware detector improves over FedAvg in specified dominant-site stress regimes and transfers without retuning from random label noise to systematic ordinal threshold shift. On CAMELYON17/WILDS, equal-client or dominant-source-adjusted weighting improves held-out-center accuracy over sample-proportional weighting in both tested feature families. TransnnMIL adds a stable custom whole-slide architecture, while the external canine and WSI-NCA experiments define important boundaries around what has and has not yet been established.

The resulting contribution is not a universal leaderboard or a clinical system. It is an end-to-end experimental framework for asking how information is represented and aggregated across scanners, tissue, slides, and institutions—and for replacing implicit assumptions about scanner invariance or sample-volume authority with explicit, testable neural and statistical mechanisms.

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